

1.SELECT:

The screenshot shows the MySQL Workbench interface with a query window titled 'Query 1 city'. The query entered is `SELECT * FROM sales;`. The 'Result Grid' displays 200 rows of data with columns: order_id, customer_name, order_date, category, sub_category, product_name, quantity, unit_price, total_price, and region. The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
41	15:32:45	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned	0.000 sec / 0.000 sec
42	15:37:45	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/sales.csv' INTO TABLE sales FIE...	Error Code: 1290. The MySQL server is running with the --secure-file-priv option so it cannot execute this state...	0.000 sec
43	15:38:18	SHOW VARIABLES LIKE 'secure_file_priv'	1 row(s) returned	0.000 sec / 0.000 sec
44	15:38:58	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.2/Uploads/sales.csv' INTO TABLE sales FIE...	200 row(s) affected Records: 200 Deleted: 0 Skipped: 0 Warnings: 0	0.016 sec
45	15:39:40	SELECT * FROM	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
46	15:39:49	SELECT * FROM sales LIMIT 0, 1000	200 row(s) returned	0.000 sec / 0.000 sec

2.WHERE:

The screenshot shows the MySQL Workbench interface with a query window titled 'Query 1 city'. The query entered is `SELECT * FROM sales WHERE region = 'West';`. The 'Result Grid' is empty. The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
45	15:39:40	SELECT * FROM	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
46	15:39:49	SELECT * FROM sales LIMIT 0, 1000	200 row(s) returned	0.000 sec / 0.000 sec
47	15:43:07	SELECT * FROM sales WHERE region = 'West' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
48	15:46:05	SELECT * FROM sales WHERE region = 'South' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
49	15:46:32	SELECT region, SUM(total_price) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	15:46:49	SELECT * FROM sales WHERE region = 'West' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec

3.GROUP BY:

The screenshot shows the MySQL Workbench interface. The 'Query 1' editor contains the following SQL query:

```
1 SELECT region, SUM(total_price) AS total_sales
2 FROM sales
3 GROUP BY region;
```

The 'Result Grid' displays the results of the query:

region	total_sales
West	599367
South	808651
East	619250
North	392839

The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
46	15:39:49	SELECT * FROM sales LIMIT 0, 1000	200 row(s) returned	0.000 sec / 0.000 sec
47	15:43:07	SELECT * FROM sales WHERE region = 'West' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
48	15:46:05	SELECT * FROM sales WHERE region = 'South' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
49	15:46:32	SELECT region, SUM(total_price) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	15:46:49	SELECT * FROM sales WHERE region = 'West' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
51	15:47:33	SELECT region, SUM(total_price) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec

4.ORDER BY:

The screenshot shows the MySQL Workbench interface. The 'Query 1' editor contains the following SQL query:

```
1 SELECT *
2 FROM sales
3 ORDER BY total_price DESC;
```

The 'Result Grid' displays the results of the query, sorted by total price in descending order:

order_id	customer_name	order_date	category	sub_category	product_name	quantity	unit_price	total_price	region
1125	Lynn Garrison	2025-05-14	Grocery	Snacks	Snacks Wish	10	4794	47940	East
1019	Debbie Turner	2025-03-22	Grocery	Snacks	Snacks Some	10	4499	44990	West
1168	Megan Charles	2025-03-26	Clothing	Jeans	Jeans Card	10	4462	44620	West
1160	Michelle Beltran	2025-03-23	Furniture	Desk	Desk Me	9	4719	42471	East
1120	Rick Sanford	2025-05-12	Furniture	Chair	Chair After	9	4579	41211	South
1070	Patrick Watson	2025-01-07	Furniture	Desk	Desk Would	8	4730	37840	West
1190	Christopher Meyer	2025-01-18	Grocery	Milk	Milk Best	8	4665	37320	East
1149	Todd Barnes	2025-02-04	Furniture	Chair	Chair Hope	8	4519	36152	North
1114	Stephanie Martinez	2025-06-01	Grocery	Milk	Milk Postion	8	4322	34576	West
1174	Kimberly Alvarado	2025-04-06	Furniture	Desk	Desk Prevent	7	4536	31752	North
1184	Gina Bush	2025-03-08	Electronics	Laptop	Laptop Enviro...	9	3466	31194	South
1167	Julie Adams	2025-03-07	Grocery	Snacks	Snacks Message	7	4444	31108	East

The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
47	15:43:07	SELECT * FROM sales WHERE region = 'West' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
48	15:46:05	SELECT * FROM sales WHERE region = 'South' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
49	15:46:32	SELECT region, SUM(total_price) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	15:46:49	SELECT * FROM sales WHERE region = 'West' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
51	15:47:33	SELECT region, SUM(total_price) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec
52	15:49:09	SELECT * FROM sales ORDER BY total_price DESC LIMIT 0, 1000	200 row(s) returned	0.000 sec / 0.000 sec

5.SUM:

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
1 SELECT customer_name, SUM(total_price) AS total_spent
2 FROM sales
3 GROUP BY customer_name
4 ORDER BY total_spent DESC
5 LIMIT 5;
```

The Result Grid shows the following data:

customer_name	total_spent
Lynn Garrison	47940
Debbie Turner	44990
Megan Charles	44620
Michelle Beltran	42471
Rick Sanford	41211

The Action Output pane shows the execution log for the query, including the time, action, message, and duration.

6.AVERAGE:

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
1 SELECT AVG(total_price) AS average_sales
2 FROM sales;
```

The Result Grid shows the following data:

average_sales
12280.5359

The Action Output pane shows the execution log for the query, including the time, action, message, and duration.

7.COUNT:

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query: `SELECT COUNT(*) AS total_orders FROM sales;`. The 'Result Grid' shows a single row with the value 200 for the column 'total_orders'. The 'Output' tab shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
49	15:46:32	SELECT region, SUM(total_price) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	15:46:49	SELECT * FROM sales WHERE region = 'West' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
51	15:47:33	SELECT region, SUM(total_price) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec
52	15:49:09	SELECT * FROM sales ORDER BY total_price DESC LIMIT 0, 1000	200 row(s) returned	0.000 sec / 0.000 sec
53	15:49:44	SELECT AVG(total_price) AS average_sales FROM sales LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
54	15:50:21	SELECT COUNT(*) AS total_orders FROM sales LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

8.JOINS:

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query: `SELECT s.customer_name, s.region FROM sales s;`. The 'Result Grid' shows a list of customer names and their regions. The 'Output' tab shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
54	15:50:21	SELECT COUNT(*) AS total_orders FROM sales LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
55	15:53:16	SELECT customer_name, SUM(total_price) AS total_spent FROM sales GROUP BY customer_name ORDER ...	5 row(s) returned	0.000 sec / 0.000 sec
56	15:53:45	SELECT AVG(total_price) AS avg_order_value FROM sales LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
57	15:54:08	SELECT customer_name, total_price, CASE WHEN total_price > 500 THEN 'High'	200 row(s) returned	0.000 sec / 0.000 sec
58	15:54:36	SELECT * FROM sales WHERE total_price > (SELECT AVG(total_price) FROM sales) LIMIT 0, 1000	81 row(s) returned	0.000 sec / 0.000 sec
59	15:55:17	SELECT s.customer_name, s.region FROM sales s LIMIT 0, 1000	200 row(s) returned	0.000 sec / 0.000 sec

9.SUB QUERIES:

The screenshot shows the MySQL Workbench interface. In the SQL editor, a query is written to select all columns from the 'sales' table where the total price is greater than the average total price of all sales. The query is as follows:

```
1 SELECT *
2 FROM sales
3 WHERE total_price > (
4     SELECT AVG(total_price)
5     FROM sales
6 )
```

The Result Grid displays the following data:

order_id	customer_name	order_date	category	sub_category	product_name	quantity	unit_price	total_price	region
1001	Misty Herrera	2025-05-05	Clothing	Jeans	Jeans New	9	1509	13581	South
1006	Breanna Gonzalez	2025-04-09	Grocery	Bread	Bread Black	6	4356	26136	East
1007	Angela Rivera	2025-06-18	Grocery	Chair	Chair Believ	7	2971	20797	West
1011	Valerie Hooper	2025-03-16	Furniture	Couch	Couch Charge	10	1759	17590	West
1018	Richard Paul	2025-05-15	Furniture	Cabinet	Cabinet Almost	6	4966	29796	North
1019	Debbie Turner	2025-03-22	Grocery	Snacks	Snacks Some	10	4499	44990	West
1020	Roy Stewart	2025-02-25	Electronics	Mobile	Mobile Charge	5	2660	13300	North
1022	Benjamin Meadows	2025-04-30	Clothing	Shoes	Shoes Job	10	2056	20560	East
1024	Jodi Smith	2025-01-30	Grocery	Bread	Bread Pass	6	4511	27066	East
1026	Kerry Moore	2025-01-09	Grocery	Bread	Bread Baby	8	2262	18096	South
1030	Ashley Joyce DVM	2025-05-11	Electronics	Headphones	Headphones Y...	9	2864	25776	South
1033	Emily Ray	2025-05-17	Electronics	Laptop	Laptop Sea	6	2564	15384	North

The Output pane shows the execution of the query, with the following messages:

```
53 15:49:44 SELECT AVG(total_price) AS average_sales FROM sales LIMIT 0, 1000 1 row(s) returned
54 15:50:21 SELECT COUNT(*) AS total_orders FROM sales LIMIT 0, 1000 1 row(s) returned
55 15:53:16 SELECT customer_name, SUM(total_price) AS total_spent FROM sales GROUP BY customer_name ORDER ... 5 row(s) returned
56 15:53:45 SELECT AVG(total_price) AS avg_order_value FROM sales LIMIT 0, 1000 1 row(s) returned
57 15:54:08 SELECT customer_name, total_price, CASE WHEN total_price > 500 THEN 'High' WH... 200 row(s) returned
58 15:54:36 SELECT * FROM sales WHERE total_price > ( SELECT AVG(total_price) FROM sales ) LIMIT 0, 1000 81 row(s) returned
```

10.CASE statements:

The screenshot shows the MySQL Workbench interface. In the SQL editor, a query is written to select all columns from the 'sales' table, categorized by total price into 'High', 'Medium', or 'Low'. The query is as follows:

```
2 total_price,
3 CASE
4     WHEN total_price > 500 THEN 'High'
5     WHEN total_price > 200 THEN 'Medium'
6     ELSE 'Low'
7 END AS sales_category
8 FROM sales;
```

The Result Grid displays the following data:

customer_name	total_price	sales_category
Caleb Davis	2394	High
Misty Herrera	13581	High
Rick Soto	5230	High
Sean Haas	2920	High
Ann Hall	1096	High
Michelle White	5458	High
Breanna Gonzalez	26136	High
Angela Rivera	20797	High
Emily Anderson	7010	High
Laura Anderson	4454	High
Ryan Smith	9834	High
Valerie Hooper	17590	High

The Output pane shows the execution of the query, with the following messages:

```
52 15:49:09 SELECT * FROM sales ORDER BY total_price DESC LIMIT 0, 1000 200 row(s) returned
53 15:49:44 SELECT AVG(total_price) AS average_sales FROM sales LIMIT 0, 1000 1 row(s) returned
54 15:50:21 SELECT COUNT(*) AS total_orders FROM sales LIMIT 0, 1000 1 row(s) returned
55 15:53:16 SELECT customer_name, SUM(total_price) AS total_spent FROM sales GROUP BY customer_name ORDER ... 5 row(s) returned
56 15:53:45 SELECT AVG(total_price) AS avg_order_value FROM sales LIMIT 0, 1000 1 row(s) returned
57 15:54:08 SELECT customer_name, total_price, CASE WHEN total_price > 500 THEN 'High' WH... 200 row(s) returned
58 15:54:36 SELECT * FROM sales WHERE total_price > ( SELECT AVG(total_price) FROM sales ) LIMIT 0, 1000 81 row(s) returned
```

11.TOP COSTUMERS:

The screenshot shows MySQL Workbench with a query window titled 'Query 1: city'. The query is as follows:

```
1 SELECT customer_name, SUM(total_price) AS total_spent
2 FROM sales
3 GROUP BY customer_name
4 ORDER BY total_spent DESC
5 LIMIT 5;
```

The result grid shows the following data:

customer_name	total_spent
Lynn Garrison	47940
Debbie Turner	44990
Megan Charles	44620
Michelle Beltran	42471
Rick Sanford	41211

The bottom panel shows the 'Action Output' for the query, listing the execution steps and their durations.

12.AVERAGE ORDER VALUES:

The screenshot shows MySQL Workbench with a query window titled 'Query 1: city'. The query is as follows:

```
1 SELECT AVG(total_price) AS avg_order_value
2 FROM sales;
```

The result grid shows the following data:

avg_order_value
12100.5350

The bottom panel shows the 'Action Output' for the query, listing the execution steps and their durations.